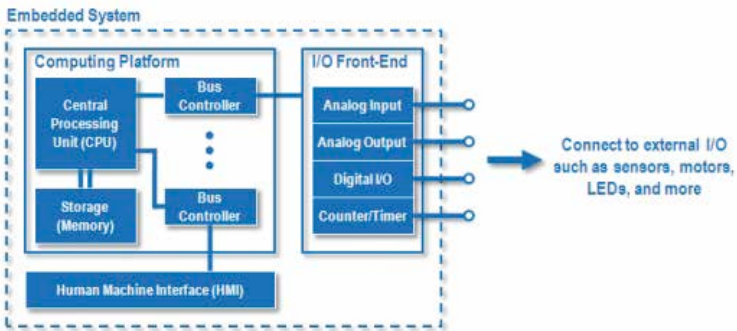


provider using their 2G/3G/4G communications radio component and can receive and send data to the Wi-Fi adapter which then can be used by the Wi-Fi adapter of another device (say, a laptop) for connecting to the internet. Both of these adapters can communicate with each other internally.

At the same time, it is necessary that the computation and communication parts of the networking scene be kept separate. The data brought in by the communication systems shouldn't be completely decoded by the communication component itself. Instead, the data is computed upon by the related processor of the embedded system. A design like this ensures that the communication gear is kept intact while the core system is changeable or vice versa. For a designer, it helps with debugging and finding problems



There can be many ways to get it done but inter-component communication is an integral part of embedded systems design

by isolating the two parts. Overall, it reduces the design complexity. Realistically, this ensures that feature updates can be sent to the software without affecting the hardware.

Security

This aspect of embedded systems, if neglected, can prove to be costly and dangerous. Most embedded systems of today are capable of establishing network connections to other systems or to the internet. Any system which transmits data should be able to communicate in both, protected and unprotected modes. This places a requirement on the networking and communication system to be able to support secure protocols. For example, a Wi-Fi transceiver should be able to communicate over open networks as well as over secured connections. However that's just one side of security.